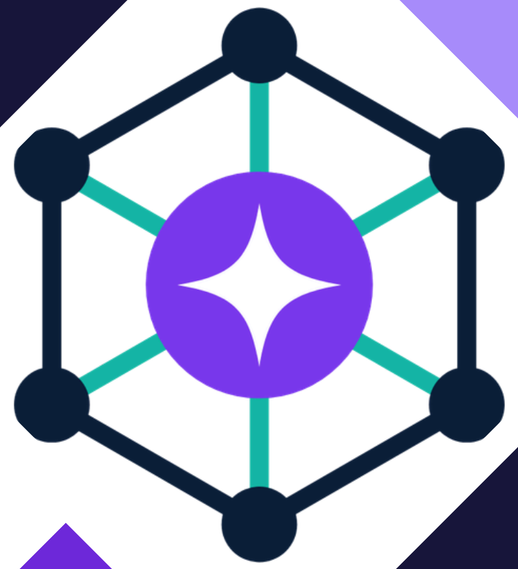


Self Network Deployer

AI-DRIVEN AUTOMATED DEPLOYMENT OF 5G TEST ENVIRONMENTS

M2: Elaboration

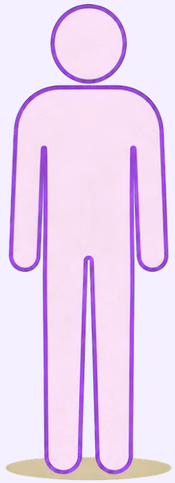
instituto de telecomunicações
universidade de aveiro
ATNOG

State of The Art

Solution	Open Source	Natural Language (Intent-to-Config)	RAG Technical Grounding	Automated Verifier Feedback Loop	5G Core Deployment	Runtime Emulation & Testing
Our project (SND)	✓	✓	✓	✓	✓	✓
Nephio	✓	✗	✗	✗	✓	✗
Cisco NSO	✗	✗	✗	✗	✓	✗
ONAP	✓	✗	✗	✗	✓	✗

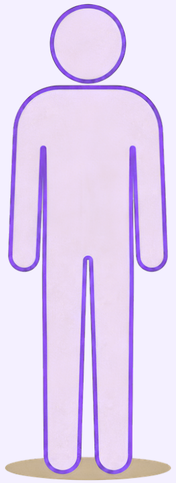
Actors

Admin



Person responsible for maintaining SND and its supported technical knowledge,

User



Anyone who uses SND to generate, save or reuse a 5G network.

Use Cases

Authentication
UC01-03
Manages user registration, login, and logout.

Running Network Management
UC10-11
Allows users to monitor the status of deployed networks and stop their execution.

Network Creation and Refinement
UC04-07
Allows users to create, review, and refine 5G networks using natural language.

Saved Network Management
UC12-14
Allows users to save, reopen, and redeploy previously created networks.

Configuration and Deployment
UC08-09
Handles the generation of network configuration files and the deployment of 5G networks.

Knowledge Base Management
UC15
Allows administrators to update and maintain the technical knowledge base used by the system.

Use Cases



Authentication

UC01

- Create Account



Running Network Management

UC10-11

- View Network Status



Network Creation and Refinement

UC04

- Create a Network from a Natural-Language Request



Saved Network Management

UC12

- Save Network Plan



Configuration and Deployment

UC08

- Generate Configuration Templates Without Deployment



Knowledge Base Management

UC15

- Update Knowledge Base

Functional-Requirements

- The system shall accept natural-language descriptions of desired 5G networks
- The system shall convert the natural-language request into a structured NetworkSpec represented in JSON
- The system shall validate the generated NetworkSpec before presenting it to the user
- The system shall allow the user to request changes to a generated or saved NetworkSpec in natural language, without editing configuration files
- The system shall generate the configuration files from a validated NetworkSpec
- The system shall deploy a validated and user-approved network to the target infrastructure
- The system shall perform runtime checks after deployment to determine whether required network components and interactions are operational
- If the runtime checks fail because of the NetworkSpec or the configuration, the system shall plan how to fix it and ask the user for confirmation before apply the fix and deploying it

Non-Functional Requirements

Security
NFR01-04
Secure credentials, authenticated communication and validated AI output

Reliability & Recovery
NFR09-13
Failure isolation, crash recovery, data integrity and safe concurrent deployments.

Usability & Performance
NFR05-08
Natural-language interaction, clear feedback and responsive UI.

Modularity & Extensability
NFR15-18
Independent components, defined interfaces and extensible plugins

Resource Management
NFR14
Controlled infrastructure allocation withing assigned resource limits and permissions.

How SND works



Front desk (AI)

Planner (AI)

User approves

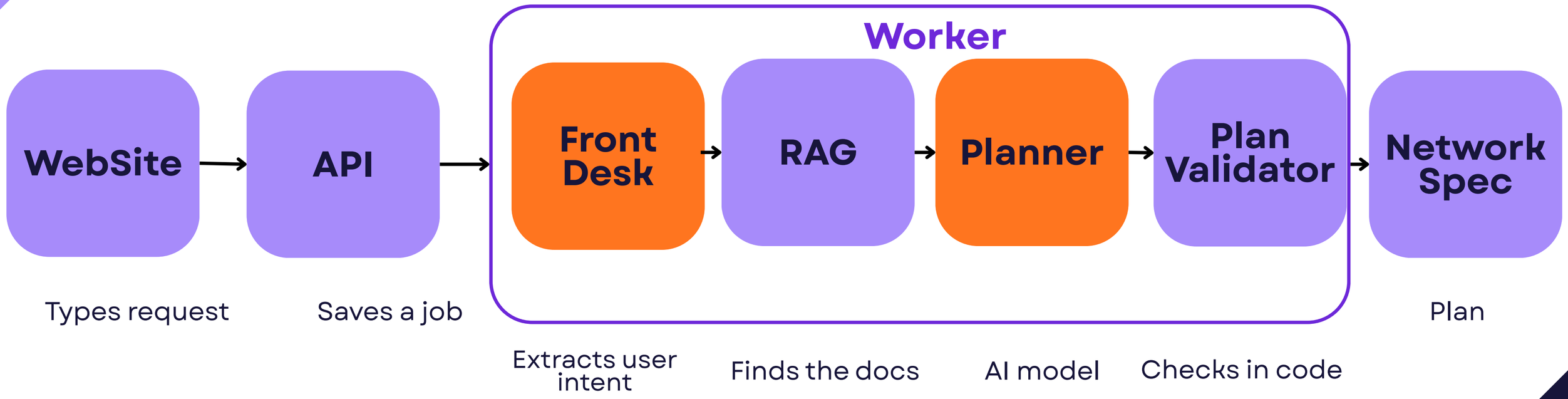
Core + plugins

Diagnosis (AI)

AI proposes, the user approves, the core executes

1. Plan

“A 5G Network with 10 UEs and 2 Slices”



2. Confirm

Here’s the plan

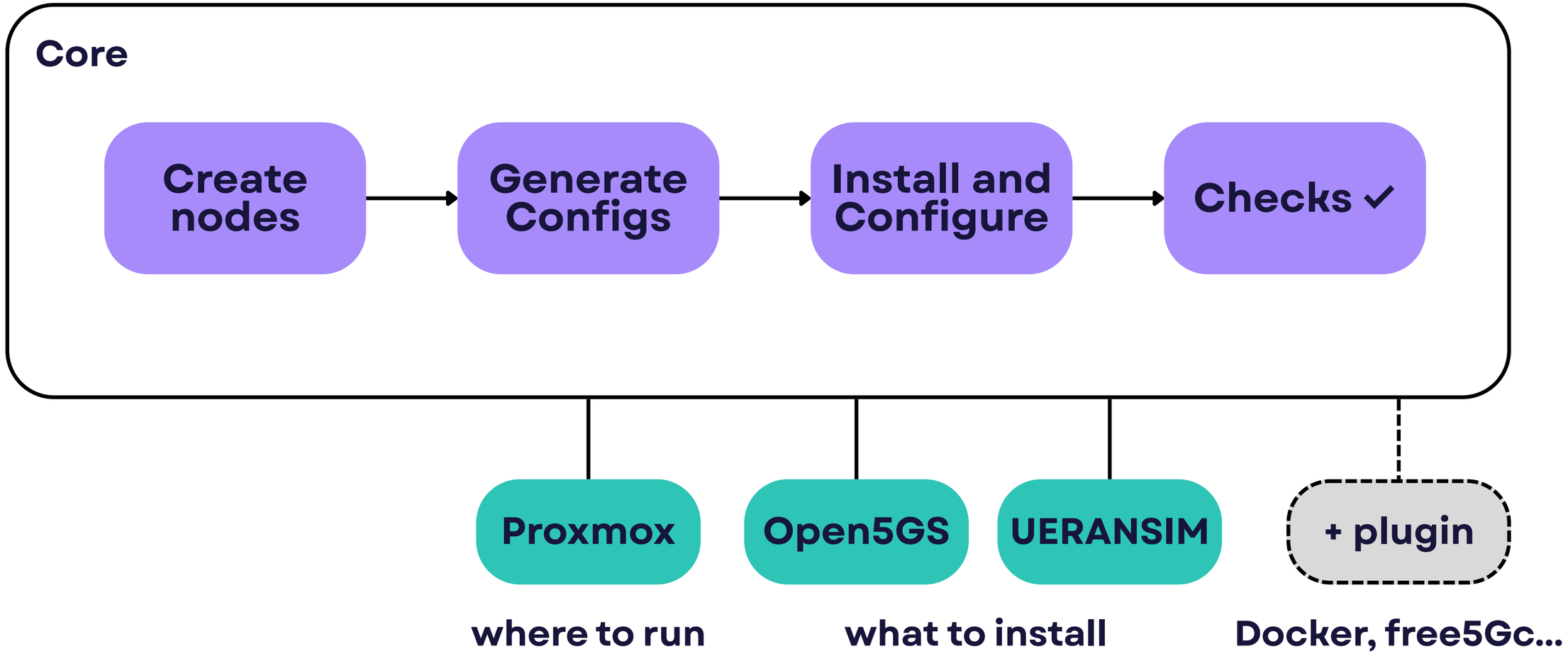
- 2 nodes (core + radio)
- 10 UE’s, 2 slices
- Needs 4 vCPU · 8 GB RAM · 2 IPs
- Free right now ✓

Deploy **Configs only** **Refine** **Cancel**

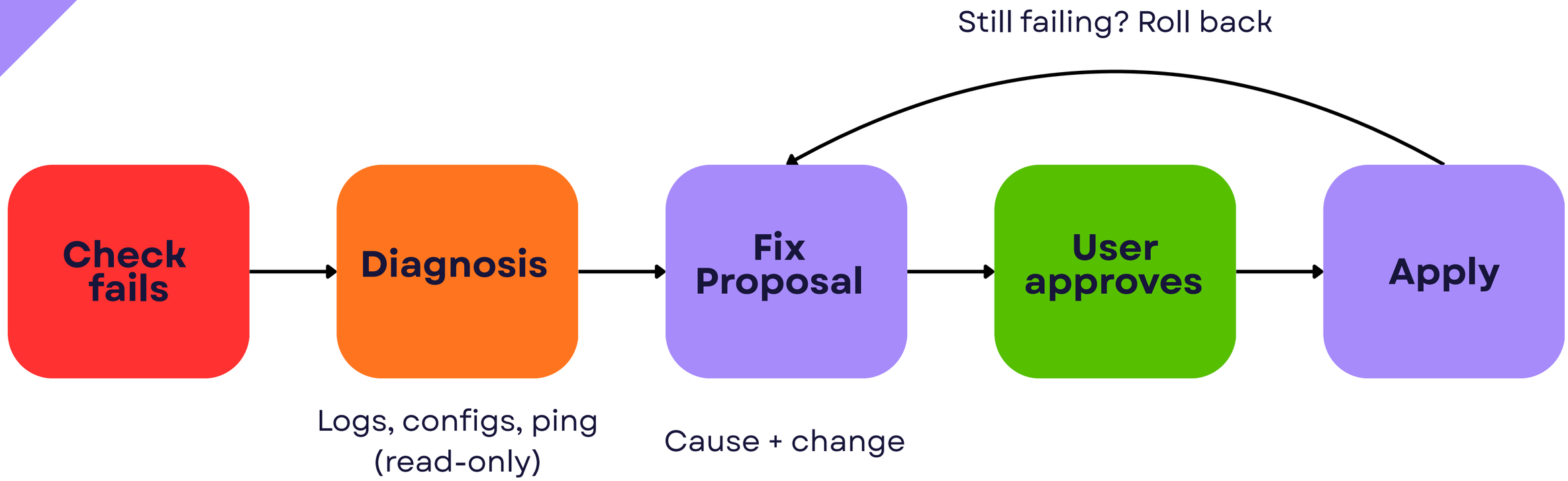
Reserve resources **New version: v1 → v2**

Nothing is built until user says yes

3.Deploy & Test

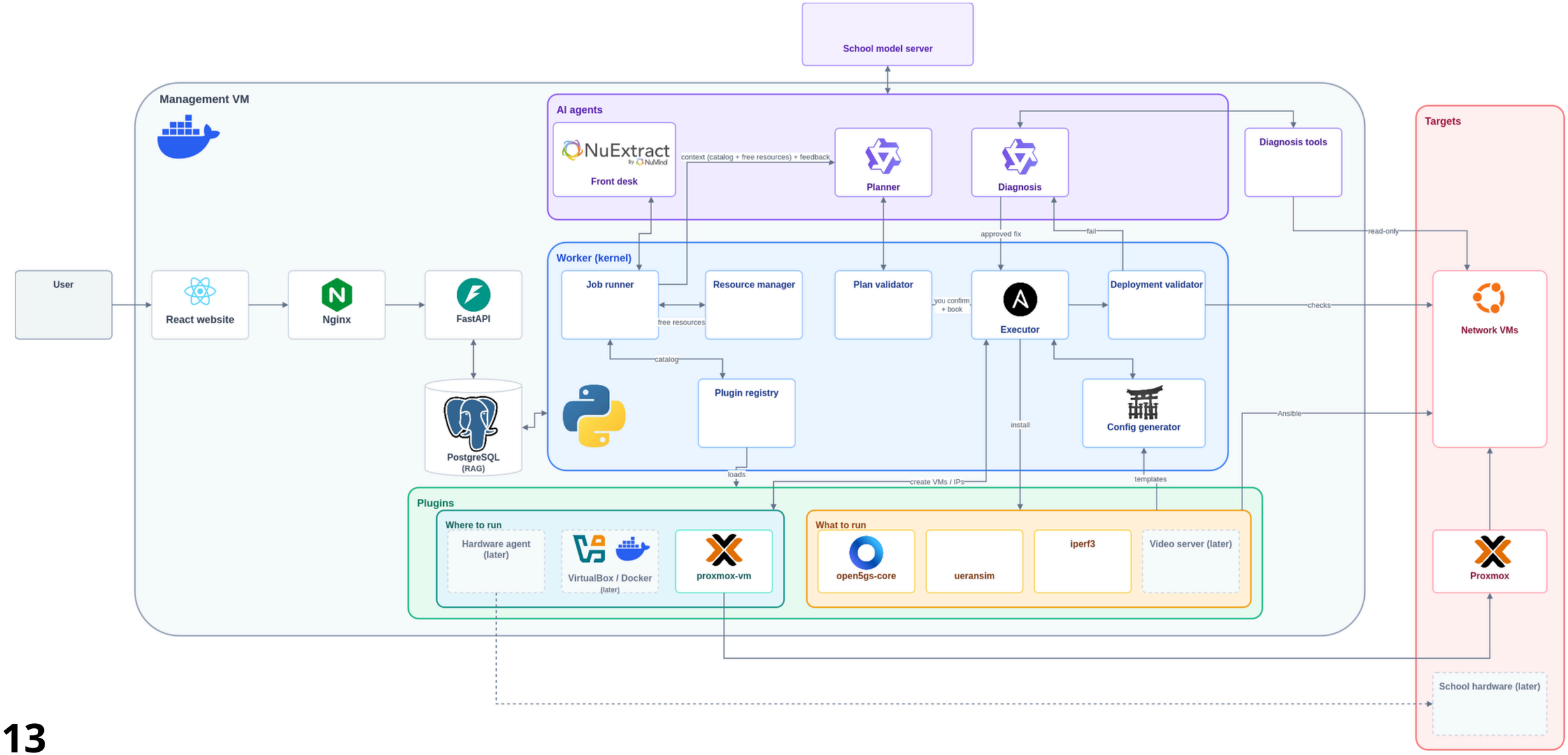


4.What if it fails?

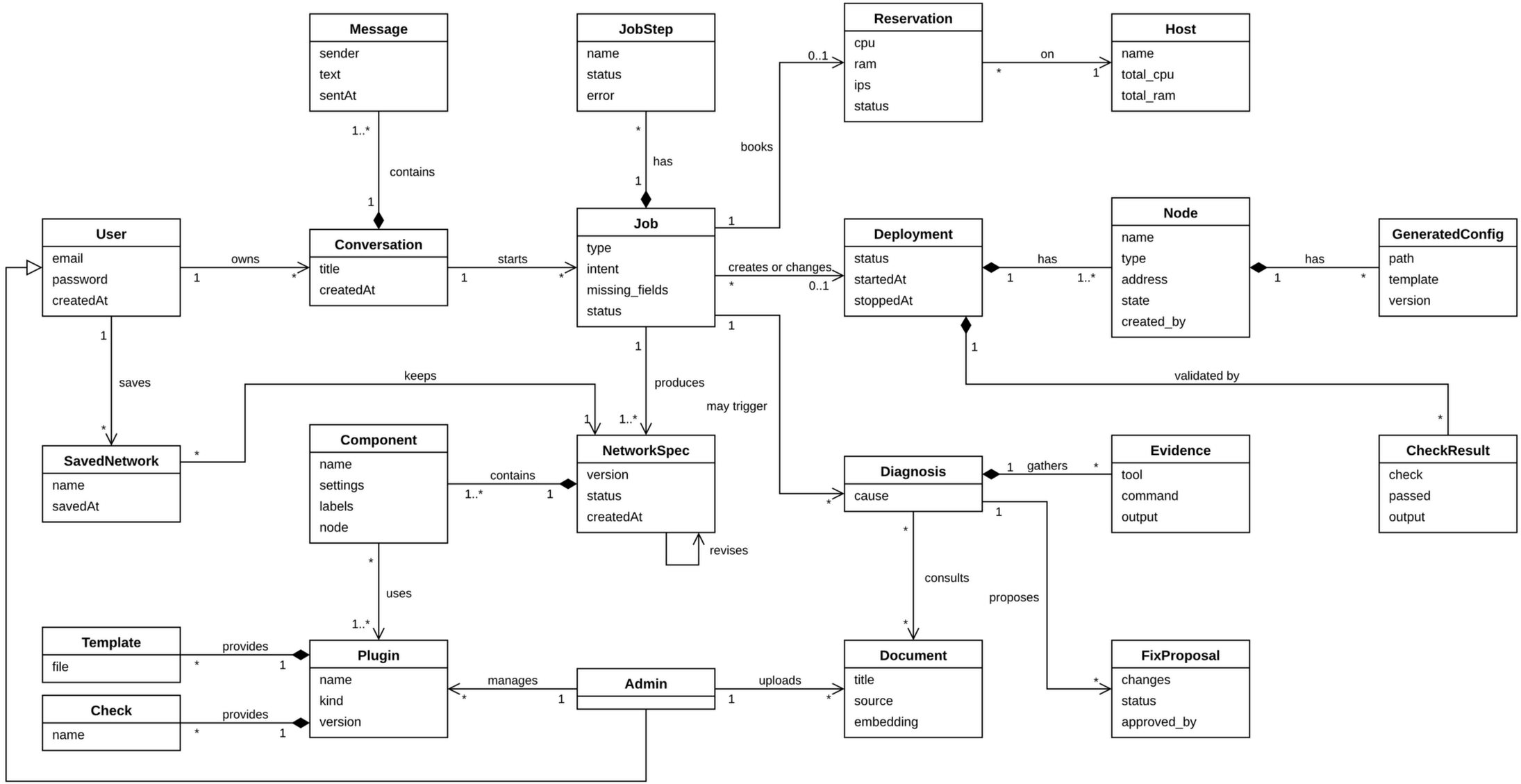


The AI investigates freely, but nothing changes without the user's approval

Architecture

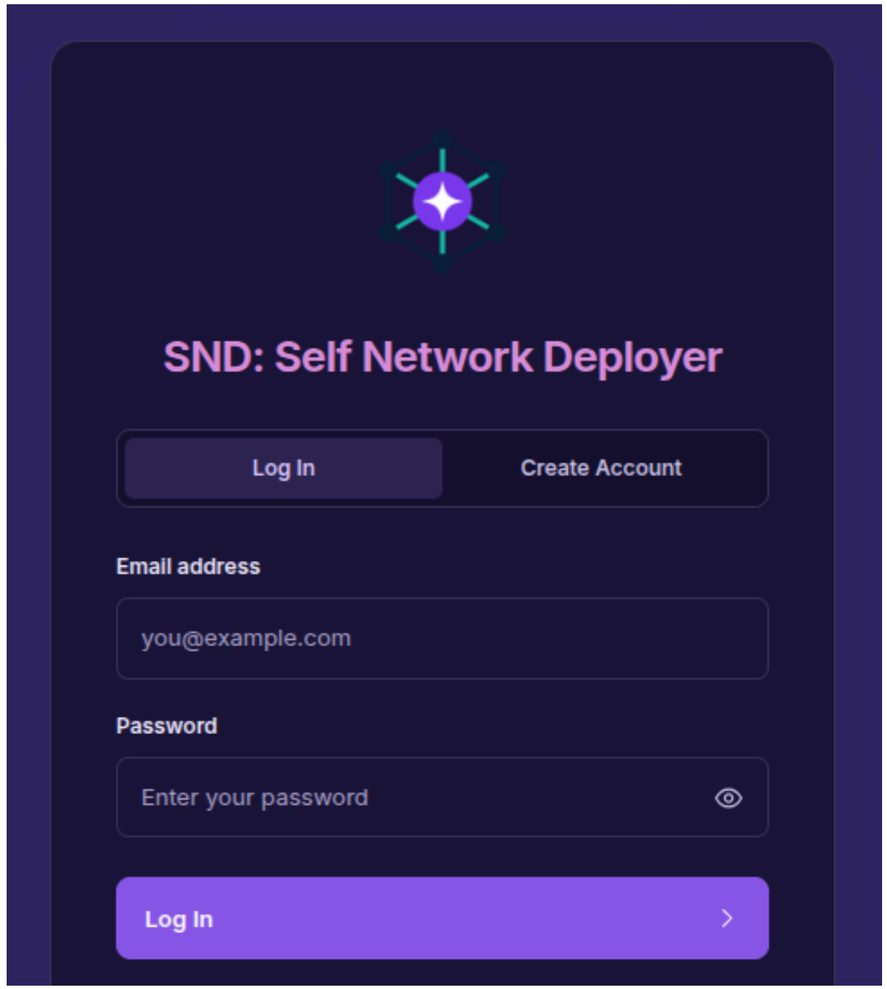


Domain Model

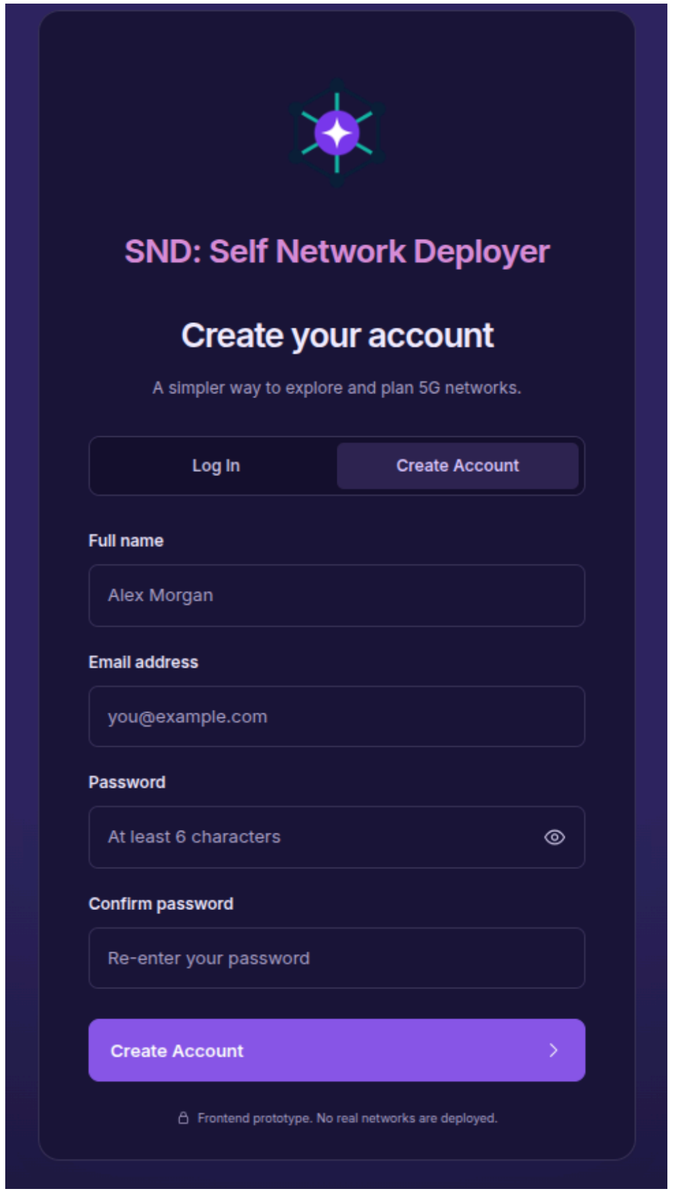


Mockups

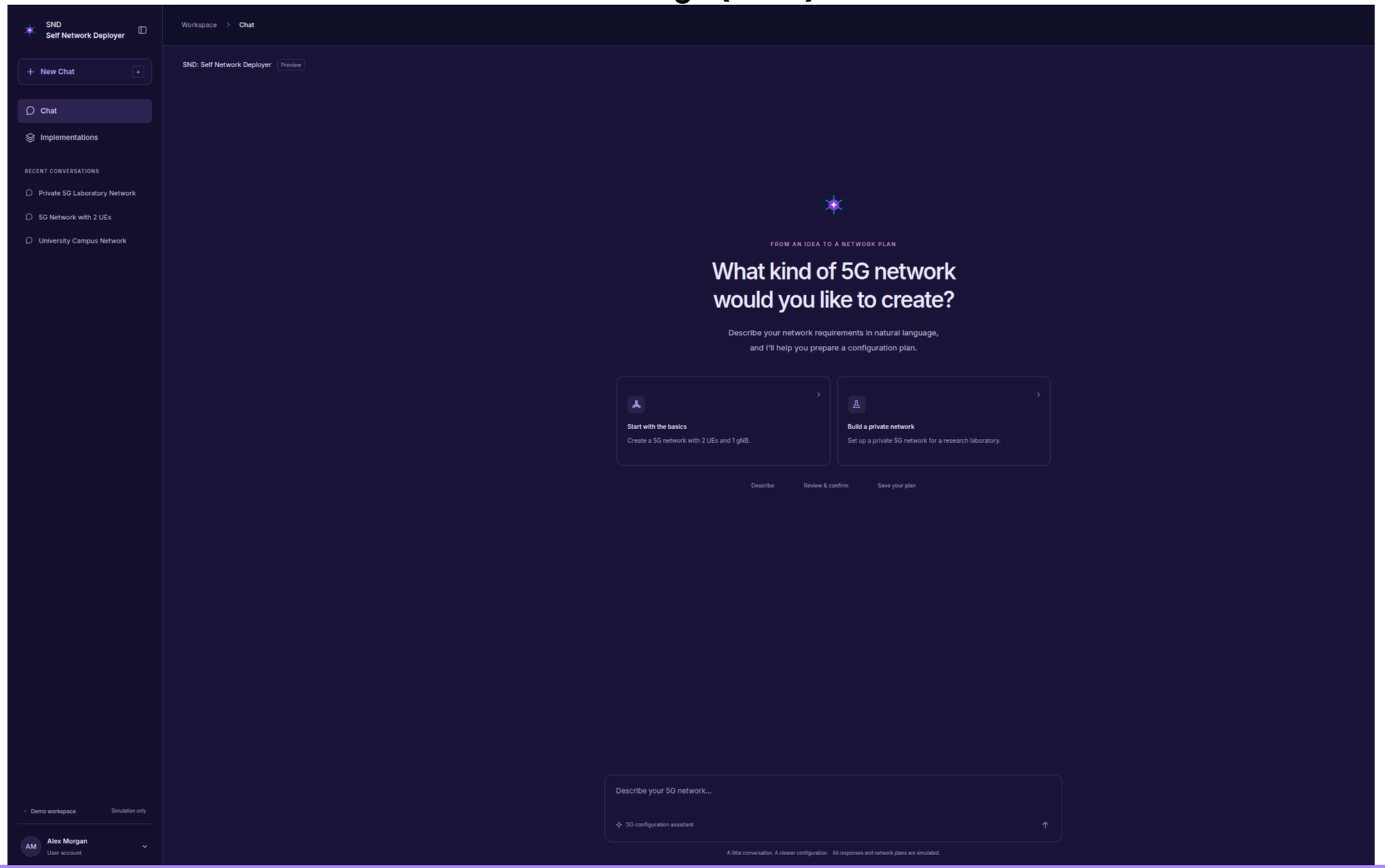
Login:



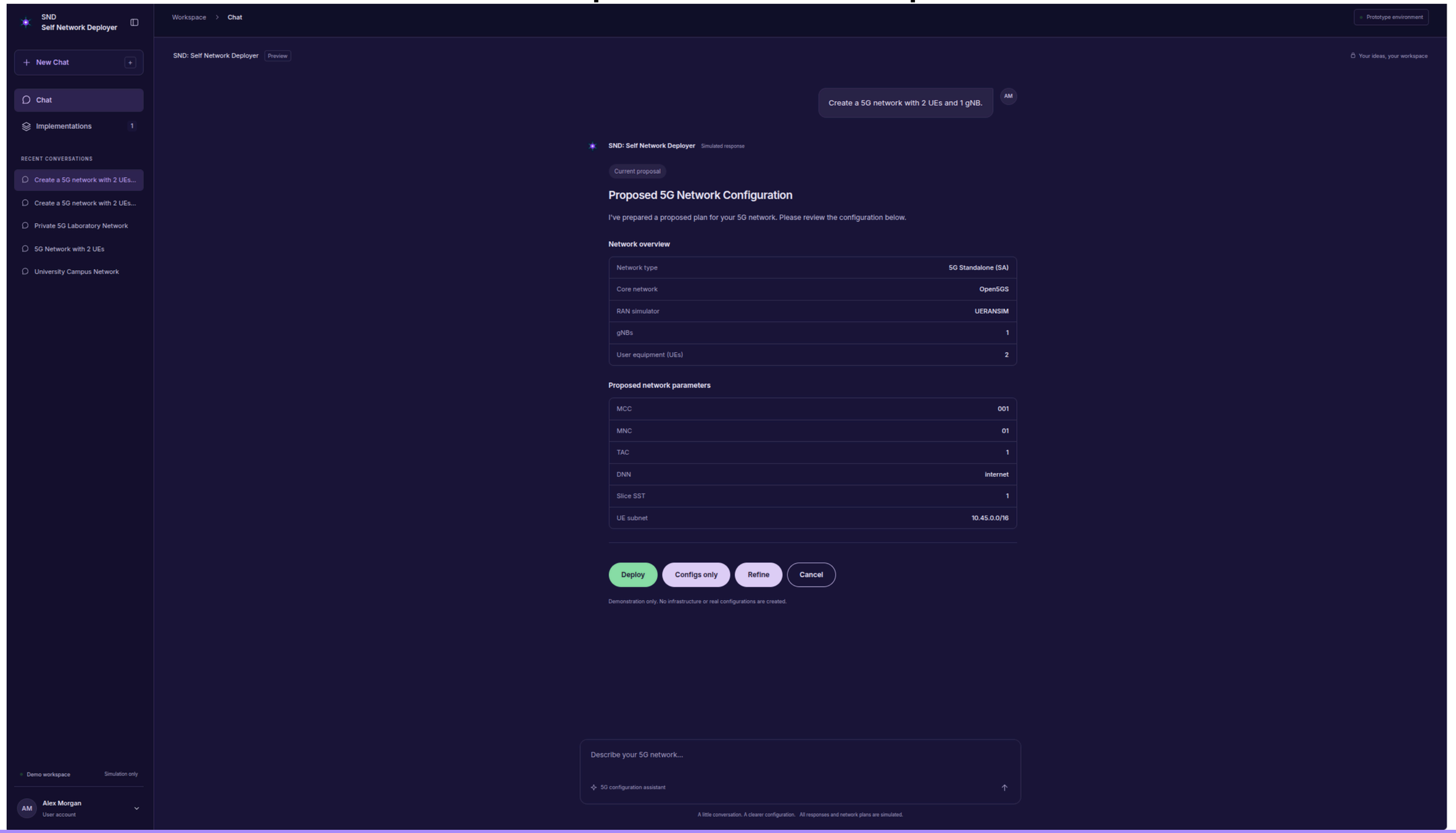
Create Account:



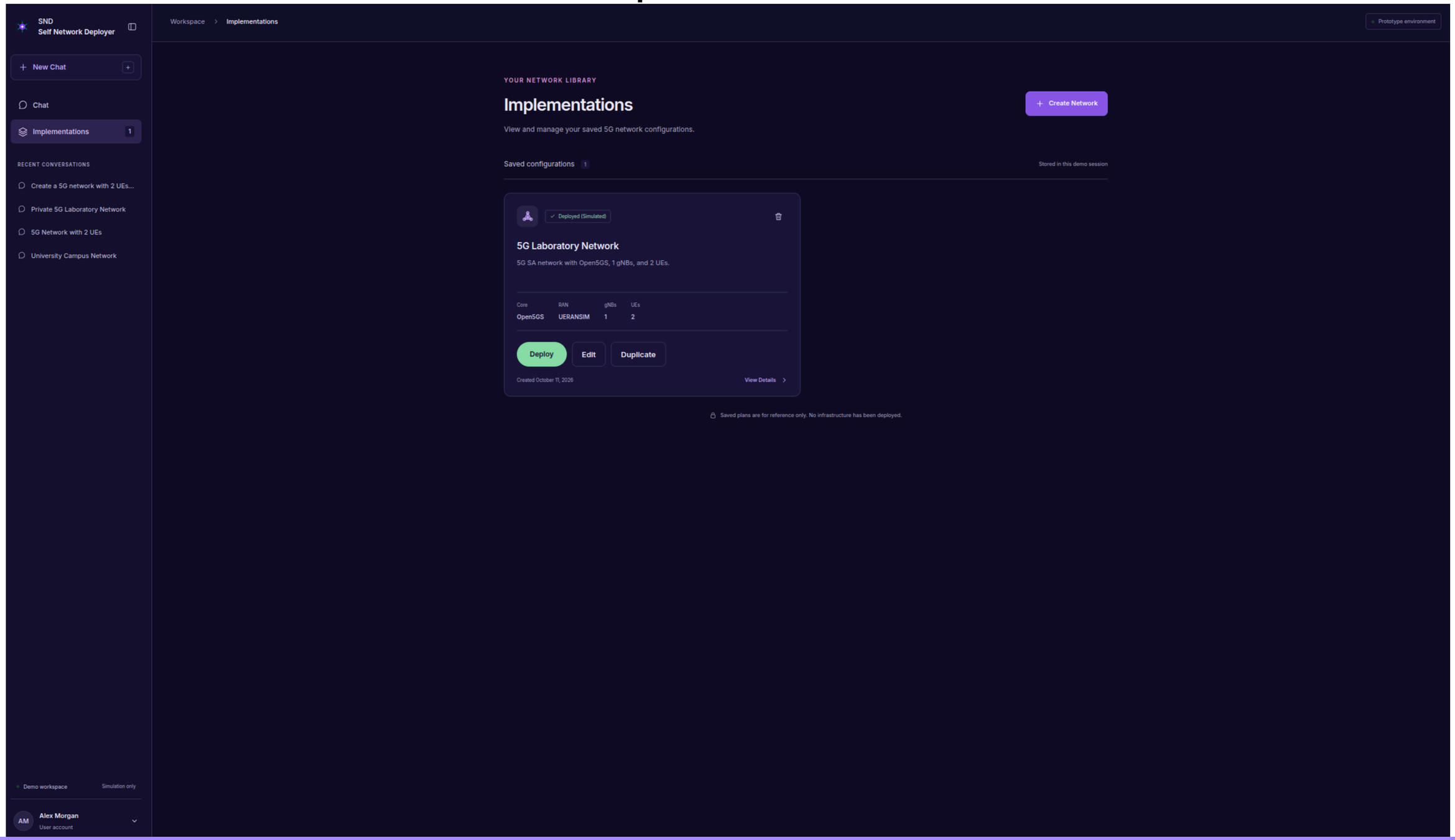
Main Page (Chat):



Network plan and feedback request :



Implementations:



References

[1] A. Clemm, L. Ciavaglia, L. Z. Granville and J. Tantsura, "Intent-Based Networking - Concepts and Definitions," RFC 9315, IRTF, Oct. 2022. https://www.rfc-editor.org/rfc/rfc9315

[2] R. Mondal, A. Tang, R. Beckett, T. Millstein and G. Varghese, "What do LLMs need to Synthesize Correct Router Configurations?," in Proc. 22nd ACM Workshop on Hot Topics in Networks (HotNets '23), 2023, pp. 189-195. https://doi.org/10.1145/3626111.3628194

[3] C. Wang, M. Scazzariello, A. Farshin, S. Ferlin, D. Kostić and M. Chiesa, "NetConfEval: Can LLMs Facilitate Network Configuration?," Proceedings of the ACM on Networking , vol. 2, no. CoNEXT2, Article 7, 2024. https://doi.org/10.1145/3656296

[4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Advances in Neural Information Processing Systems 33 (NeurIPS 2020), 2020, pp. 9459-9474. https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

[5] A.-L. Bornea, F. Ayed, A. De Domenico, N. Piovesan and A. Maatouk, "Telco-RAG: Navigating the Challenges of Retrieval Augmented Language Models for Telecommunications," in Proc. IEEE Global Communications Conference (GLOBECOM 2024), 2024. https://doi.org/10.1109/GLOBECOM52923.2024.10901158

[6] İ. E. Sarıdaş, O. Salan, A. Görçin, İ. Hökelek and H. A. Çırpan, "RAG-Driven Multi-Agent LLM Framework with Task Decomposition for Beyond 5G Auto-Configuration," in Proc. 32nd International Conference on Telecommunications (ICT), 2026. https://doi.org/10.1109/ICT70370.2026.11594648

[7] M. Bjorklund (Ed.), "The YANG 1.1 Data Modeling Language," RFC 7950, IETF, Aug. 2016. https://www.rfc-editor.org/rfc/rfc7950

[8] OASIS, "TOSCA Version 2.0," OASIS Standard, Jul. 2025. https://docs.oasis-open.org/tosca/TOSCA/v2.0/TOSCA-v2.0.html

References

[9] Nephio Project, "Nephio Documentation." https://docs.nephio.org/docs/

[10] Cisco Systems, "NSO at a Glance," Cisco Crosswork NSO Documentation. https://nso-docs.cisco.com/nso-basics/nso-at-a-glance

[11] Cisco Systems, "NSO MCP Server," Cisco Crosswork NSO Documentation. https://nso-docs.cisco.com/guides/development/core-concepts/northbound-apis/nso-mcp-server

[12] ONAP Project, "ONAP Architecture Overview," ONAP Documentation. https://docs.onap.org/en/latest/ecosystem/architecture/index.html

[13] Open5GS, "Open5GS: Open Source implementation for 5G Core and EPC." https://open5gs.org/

[14] UERANSIM, "UERANSIM: Open source 5G UE and RAN (gNodeB) implementation" GitHub. https://github.com/aligungr/UERANSIM

[15] S. Herle, "docker_open5gs," GitHub. https://github.com/herlesupreeth/docker_open5gs

Questions?



For more information:

https://mariosantoslp.github.io/SND-microsite/